Akoizo Original Practice - Form 5 - Answer key

Division B | Scrambler | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which egg holder detail is NOT legal?

Unit: Protecting the egg.
Worked-case data: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.
Separate investigation observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

A. A balsa backstop

B. A 3/8 inch dowel

C. Dowels 3.5 cm long

D. Dowel bottoms 7 cm above the track

Correct: A

2. (2 points) What happens if string piles up on the axle?

Unit: Drivetrains: turning a drop into distance.
Worked-case data: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?
Separate investigation observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

A. Effective axle radius grows, changing distance per centimeter of string

B. Friction disappears

C. The wheel shrinks

D. Nothing

Correct: A

3. (2 points) Which is the only allowed source of propulsion energy?

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

A. A falling mass up to 2.00 kg

B. A wound rubber band

C. A battery motor

D. A compressed spring

Correct: A

4. (2 points) When is the Can Bonus lost?

Unit: The Can Bonus and curved paths.
Worked-case data: Target 9.00 m. The team sets a 60 cm gap. Find the bonus, the gap center offset, and the arc radius.
Separate investigation observations:
Arc runs: Passed between cans 10 of 10.
Stopping: Stopped 18 cm short of the target each time.
Setting: Turns calculated from the straight-line distance.

A. If the egg is foremost

B. If the gap is over 50 cm

C. If any part of the vehicle does not pass between the cans or a can is moved

D. If the run takes over 5 s

Correct: C

5. (4 points) Explain how a wing nut mechanism sets distance.

Unit: Automatic stopping.
Worked-case data: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?
Separate investigation observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

answer: As the axle turns, the nut travels along a threaded rod; after a set number of turns it triggers the brake. (4 points).

6. (4 points) Why is consistent release important?

Unit: Energy from a falling mass.
Worked-case data: A 1.50 kg mass falls 0.60 m. If 40 percent reaches a 0.50 kg vehicle as kinetic energy, what is its top speed?
Separate investigation observations:
2.0 kg mass: Top speed 4.2 m/s; egg cracked on stop in 3 of 5 runs.
1.2 kg mass: Top speed 2.6 m/s; egg safe 5 of 5; time +1.8 s.
Target: 8.00 m in both cases.

answer: The same energy each run makes stopping position and calibration predictable. (4 points).

7. (4 points) Why may participants only touch the pencil during triggering?

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

answer: Touching the vehicle could add energy or aim it, which the rules prohibit. (4 points).

8. (4 points) What violations involve the Terminal Barrier?

Unit: Protecting the egg.
Worked-case data: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.
Separate investigation observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

answer: Any part besides the egg touching it, or the egg tip passing its front. (4 points).

9. (4 points) Why release the string when fully unwound?

Unit: Drivetrains: turning a drop into distance.
Worked-case data: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?
Separate investigation observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

answer: So the axle does not rewind the string or snag, which could stop or reverse the vehicle. (4 points).

10. (6 points) Solve this case and explain your reasoning: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.

reasoning: Distance Score = 2.0 x 12.4 = 24.80; Time Score = 5.31.
Can Bonus = -0.5 x (110 - 60.0) = -25.00.
Run Score = 100 + 24.80 + 5.31 - 25.00 = 105.11. (3 points).

conclusion: Run Score = 105.11. (3 points).

11. (6 points) Solve this case and explain your reasoning: Target 9.00 m. The team sets a 60 cm gap. Find the bonus, the gap center offset, and the arc radius.

reasoning: Bonus = -0.5 x (110 - 60) = -25 points.
Gap spans 40 to 100 cm left of center; center offset s = 0.70 m.
R = (9.00^2/4 + 0.70^2) ÷ (2 x 0.70) = (20.25 + 0.49) ÷ 1.40 about 14.8 m. (3 points).

conclusion: Bonus -25 with an arc radius near 14.8 m. (3 points).

12. (2 points) About what setting should the team use?

Supplied synthetic observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

A. About 33.3 turns

B. 32.0 turns

C. 35.0 turns

Correct: A

13. (2 points) Define “Gravitational potential energy” in the context of Scrambler.

answer: mgh released as the mass falls. (2 points).

14. (2 points) Define “Target Distance” in the context of Scrambler.

answer: 7.00-10.00 m, announced after impound. (2 points).

15. (2 points) Define “Dowels” in the context of Scrambler.

answer: Two wooden pegs supporting the egg. (2 points).

16. (2 points) Define “Wheel circumference” in the context of Scrambler.

answer: 2πR, distance per wheel turn. (2 points).

17. (2 points) Define “Distance Score” in the context of Scrambler.

answer: 2.0 points per cm of Vehicle Distance. (2 points).

18. (2 points) Define “Inside Can Distance” in the context of Scrambler.

answer: Gap between cans used in the bonus formula. (2 points).